

TAUFIQUL HAQUE KHAN TUSAR

AI/ML RESEARCHER · AI AND BACKEND ENGINEER

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Education

Bachelor of Science (B.Sc.) in Computer Science & Engineering

Apr 2018 – Nov 2022

City University

Dhaka 1216, Bangladesh

- **Thesis:** “Automated Detection of Acute Lymphoblastic Leukemia Subtypes from Microscopic Blood Smear Images Using Deep Neural Networks,” doi: [10.48550/arXiv.2208.08992](https://doi.org/10.48550/arXiv.2208.08992)
- **Research Focus:** Natural Language Processing, Healthcare AI, and Accessible AI.
- **Research Achievements:** 4 peer-reviewed publications (1 journal, 3 conferences).
- **CGPA:** 3.21 / 4.00

Publications

Journal: 1 • **Conference:** 3 • **Pre-print:** 1 • **Citations:** 180+ as of August 2026 • **h-index:** 5

- [1] **Taufiqul Haque Khan Tusar** et al., “An Intelligent Telediagnosis of Acute Lymphoblastic Leukemia using Histopathological Deep Learning,” Journal of Computing Theories and Applications (JCTA), vol. 2(1), pp. 1–12, 2024. doi: [10.62411/jcta.10358](https://doi.org/10.62411/jcta.10358).
- [2] J. Hossain, M. T. Islam, and **Taufiqul Haque Khan Tusar** “Streamlining Brain Tumor Classification with Custom Transfer Learning in MRI Images,” 2023 IEEE SIST, Astana, Kazakhstan, 2023, pp. 522-526, doi: [10.1109/SIST58284.2023.10223507](https://doi.org/10.1109/SIST58284.2023.10223507).
- [3] **Taufiqul Haque Khan Tusar**, M. T. Islam, and A. H. Sakil, “An experimental study for early diagnosing Parkinson’s disease using machine learning,” arXiv preprint, 2023. doi: [10.48550/arXiv.2310.13654](https://doi.org/10.48550/arXiv.2310.13654).
- [4] **Taufiqul Haque Khan Tusar**, M. T. Islam and F. I. Raju, “Detecting Chronic Kidney Disease (CKD) at the Initial Stage: A Novel Hybrid Feature-selection Method and Robust Data Preparation Pipeline for Different ML Techniques,” 2022 5th IEEE ICCI, Cairo, Egypt, 2022, pp. 400-407, doi: [10.1109/ICCI54321.2022.9756094](https://doi.org/10.1109/ICCI54321.2022.9756094).
- [5] **Taufiqul Haque Khan Tusar** and M. T. Islam, “A Comparative Study of Sentiment Analysis Using NLP and Different ML Techniques on US Airline Twitter Data,” 2021 IEEE ICECIT, Khulna, Bangladesh, 2021, pp. 1-4, doi: [10.1109/ICECIT54077.2021.9641336](https://doi.org/10.1109/ICECIT54077.2021.9641336).

Research Experience

Research Engineer

Feb 2023 - Mar 2023

AIMS Lab, United International University

Dhaka 1212, Bangladesh

- Implemented the referenced research work (doi.org/10.1109/ETCCE54784.2021.9689831) into a real-world application.
- Developed the backend for a multi-modal AI receptionist integrating face, voice, and speaker recognition.

Undergraduate Research Assistant

Dhaka 1216, Bangladesh

City University (Supervisor: Md. Touhidul Islam)

Dec 2020 - Nov 2022

- Contributed to machine learning research in healthcare (leukemia, Parkinson’s, CKD) and NLP applications.
- Co-authored four peer-reviewed publications; Trained and fine-tuned deep learning models.
- Deployed two machine learning models into serviceable Flask applications.

Industry Experience

Independent Product Engineer

Feb 2026 - Present

ChatCopilot.app

Gazipur 1750, Bangladesh

- Architected and built ChatCopilot.app, a multi-tenant Django SaaS for AI-assisted sales, customer support, product discovery, orders, and appointment bookings.
- Implemented an agent runtime with tool calling, conversation state, history compaction, model routing, Bengali translation, image understanding, audio transcription, and human handover.
- Built tenant-scoped RAG for business knowledge and product/service catalogs using pgvector and multimodal embeddings.
- Designed integrations for Meta Graph API, website chat widget, Google Calendar, custom business ERP calendars, outbound webhooks, and Pathao courier dispatch.
- Implemented subscription plans, per-tenant quotas, LLM usage and cost accounting, sandbox testing, asynchronous message batching, and operational dashboards.
- Managed the DigitalOcean production droplet, production monitoring, resource usage, and PostgreSQL database.

AI and Application Engineer

KafKai Giken Inc.

May 2023 - present

Tokyo 110-0016, Japan (Remote)

- Built and maintained the Django-PostgreSQL service and AI agents of [KAFKAI V3](#).
- Reduced LLM API costs by 50% by offloading/routing to task-specific capable models and providers.
- Built the FastMCP server, implemented action-based credit usage, and Stripe top-up for the upcoming Kafkai V4.
- Fine-tuned and maintained in-house LLMs from HuggingFace for long-running tasks, API cost-reduction, and data safety.
- Maintained the legacy Pyramid-based codebase of Kafkai V2 and converted it to Django-PostgreSQL architecture.
- Managed the deployment, monitoring, sentry, kanban, and technical support of multiple SaaS products.
- Collaborated with UI/UX team to improve user experience of [Kafkai.com](#), [Kafkai.ai](#), and [Puchiden.app](#).
- Conducted product cost and revenue analysis, pricing strategies, and competitor studies for [Kafkai.com](#) and [KaiMail.net](#).
- Managed the affiliate and partner programs of [Puchiden.app](#).
- Conducted technical interviews and the hiring process for the Engineering team.

Machine Learning Engineer (Intern)

Shadhin Lab LLC

July 2022 - Sept 2022

Dhaka 1219, Bangladesh

- Developed NLP-based large volume data collection, labeling, and classification pipelines for an in-house product.

Teaching Experience

Assistant Teacher

Milton English & Math Academy

Gazipur 1750, Bangladesh

Nov 2019 - Mar 2020

- Taught Physics and Information & Communication Technology to secondary and higher-secondary students.
- Mentored students in fundamental computational and problem-solving concepts.

Academic & Technical Services

- **Reviewer**, EuroPython 2025, Prague, Czechia (14-20 July 2025)
- **Reviewer**, 3rd International Conference on Computing and Information Technology (ICCIT), 2023, Tabuk, Saudi Arabia
- **Technical Committee Member**, International Conference on Electronics, Information and Industrial Engineering, 2024, Wuhan, China.

Conferences

- **Speaker**, PyCon ID 2025, Jakarta, Indonesia (14-15 December 2025)
- **Presenter**, 5th International Conference on Computing and Informatics (ICCI), 2022, Cairo, Egypt
- **Presenter**, International Conference on Electronics, Communications and Information Technology, 2021, Khulna, Bangladesh

Achievements

- **Best Paper Award**, 2023 IEEE International Conference on Smart Information System and Technologies
- **Merit Scholarship**, City University, Bangladesh — Summer 2020
- **Junior School Government Scholarship**, Ministry of Education, Bangladesh — 2012

Technical Skills

Languages	Python, SQL, Bash, C
GenAI	Transformers, PyTorch, peft, LangGraph, FastMCP, llama-server, pgvector, Chroma
LLM APIs	OpenAI, Anthropic, OpenRouter, Perplexity, JinaAI, Ollama
ML & Data	scikit-learn, NLTK, OpenCV, pandas, NumPy, SciPy, Matplotlib
Backend	Django, FastAPI, Flask, Wagtail, Celery, Redis, Pytest, PostgreSQL, SQLite, Stripe
Deployment	AWS (EC2, Lightsail, S3), DigitalOcean, Render, Cloudflare (R2, Workers, D1, etc.), Caddy, Nginx
Others	REST APIs, OAuth, Webhooks, Git & GitHub, Linux, LaTeX, HTML, HTMX, Ajax, Jira/ClickUp

Technical Projects

- **MochaChat**  
Multi-character AI companion with streaming conversations. Built stateless FastAPI backend with browser-local per-character memory, streaming responses, LLM memory compaction, multilingual support, and cheap LLM routing via OpenRouter.
Tools: FastAPI, OpenRouter, Render

- **Python Buddy** 🔄

Fine-tuned a small language model (Qwen2.5-coder-1.5-instruct) for code assistance, worked on low-resource training and inference. Achieved 7–12 tokens/s inference speed with 8-bit quantization on Snapdragon 685.

Tools: HuggingFace Transformers, Llama.cpp, Python

- **Inference API Server** 🔄

Developed a lightweight Python API server for local/self-hosted LLM inference. Supports GGUF models with both string-based and role-based prompts, returns OpenAI-compatible responses.

Tools: Python, HuggingFace Hub, Llama-cpp-Python

- **Bengali AI Receptionist** 🔄

Developed an AI receptionist Flask app integrating face recognition, speech-to-text, and text-to-speech. Verifies registered users and answers user queries using semantic similarity search.

Tools: OpenCV, Sentence-Transformers, Flask, gTTS

- **Blood Cancer Stage Detection App** 🔄

Built a deep learning web app for acute lymphoblastic leukemia (ALL) stage detection. Finetuned and evaluated 4 CNN architectures; MobileNetV2 achieved 98% accuracy across 3 leukemia subtypes.

Tools: TensorFlow, Flask, OpenCV, Gunicorn, Render

- **Chronic Kidney Disease Diagnosis App** 🔄

Built a Flask-based ML app for early chronic kidney disease detection. Used LOF for outlier removal, SMOTE for imbalance handling, and Random Forest achieved maximum accuracy on test data.

Tools: Scikit-learn, Flask, Gunicorn, Joblib

Community & Outreach

Community Lead

March 2025 - Present

AI/ML Professional Community Bangladesh

Bangladesh

- Organize technical webinars and workshops on AI/ML applications.
- Mentor early-career practitioners in development and career pathways.

Community Organizer

Jan 2025 - Present

Python Bangladesh

Bangladesh

- Organize tech meetups and collaborative events for Python developers across Bangladesh.
- Lead initiatives to restart PyCon Bangladesh.

Artificial Intelligence Mentor

June 2022 - March 2023

The Empty Brain

Dhaka 1212, Bangladesh

- Led technical teams in a startup incubator, ranked at top 25 in StartKoro Season 3.
- Mentored 20+ undergraduates in python, machine learning, and research methodologies.

JR. Machine Learning Engineer

Dec 2022 - Feb 2023

Omdena

CA 94301, USA (Remote)

- Enhanced a real-time Smart Traffic system with computer vision to reduce congestion on urban motorways.
- Collaborated with a global team (50+ members) to collect and process traffic image data.

Data Science Mentor

Sept 2021 - Oct 2021

AI Planet (Formerly DPHI)

Hyderabad, India (Remote)

- Supported the core team in running the data science boot-camp.
- Guided participants on Exploratory Data Analysis, created and evaluated final assessments.

Memberships

- **Member**, IEEE Computer Society (Aug 2022 - Dec 2022)
- **Student Member**, Institute of Electrical and Electronics Engineers (IEEE) (Aug 2021 - Dec 2022)
- **Executive Member**, City University Computer Club (Dec 2020 - Oct 2022)

Extracurricular Activities

- **Student Ambassador**, IEEEExtreme (Aug 2022 - Oct 2022)
- **Chapter Influencer**, Google Crowdsourcing (Sep 2021 - Jun 2022)
- **V-Force Volunteer**, United Nations Volunteer (Aug 2021 - Sep 2021)

References

- **Kamal Mustafa** — ✉ kamal@kafkai.com
CTO, Kafkai Giken Inc., Tokyo 110-0016, Japan
- **Md. Touhidul Islam** — ✉ touhid000cse@gmail.com
Sr. Software Engineer, Samsung R&D Institute, Dhaka, Bangladesh
Former Lecturer, Dept. of CSE, City University, Dhaka 1216, Bangladesh
- **Brig. General (Rtd.) Prof. Dr. Engr. Md. Lutfor Rahman** — ✉ lutfor.mist@yahoo.com
Dean, Faculty of Science & Engineering, City University, Dhaka 1216, Bangladesh
- **Dr. Rubens Talukder** — ✉ nexeltinc@gmail.com
Sr. Program Manager (North America), Tata Consultancy Services, MI, USA